



The
University of
Faisalabad

Firewalls

Lecture 19



Definition?

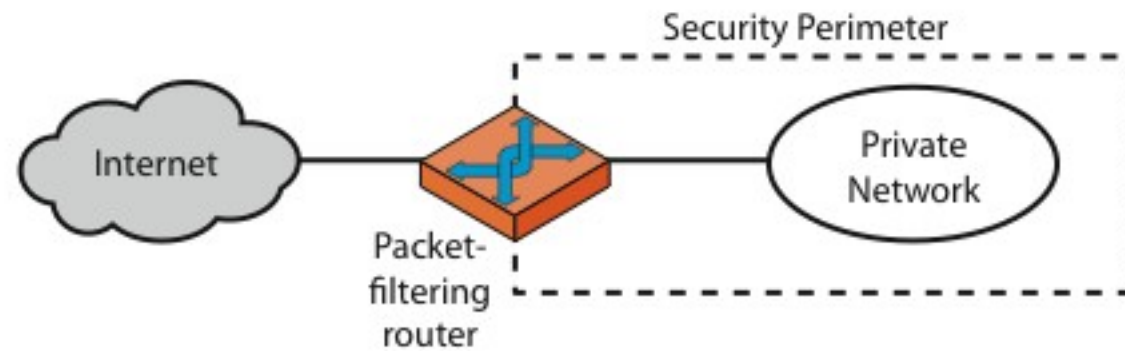
- A **firewall** is a security system that acts as a **protective barrier** between different networks, like your private network and the internet. It works as a **checkpoint** that controls and monitors data traffic, allowing only **authorized access** while blocking harmful or unauthorized activities.
- **Key Features in Simple Terms:**
- **Traffic Control:** It decides what data can pass through and what should be blocked.
- **Security Barrier:** Protects trusted networks (like your home or office) from untrusted ones (like the internet).
- **Access Restrictions:** Only allows approved users and services.

- **Monitoring & Alerts:** Keeps logs and alerts for suspicious activities.
- **Defense Against Attacks:** It is designed to be secure and not easily hacked.
- **Perimeter Protection:** Like a security guard at the entrance of a building, it helps keep the network safe.

Characterized by protocol level it controls in

- Packet filtering
 - Circuit gateways
 - Application gateways
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- Combination of above is dynamic packet filter

Packet Filter



(a) Packet-filtering router

How it Works:

- A data packet comes from the **Internet**.
- The **Packet-Filtering Router** checks its details (source, destination, protocol, etc.).
- If it follows security rules, it enters the **Private Network**; otherwise, it's blocked.
- **Example:**
-  **Allowed:** A request from a trusted website (Google.com) is permitted.
-  **Blocked:** A hacker trying to access a restricted port is denied.
- This firewall type is simple and efficient but may not catch sophisticated attacks like malware.

- A **packet-filtering firewall** is the most basic type of firewall that checks data packets based only on a few key details. It **does NOT** inspect the actual content of the data—just the "labels" on the packet.
- **What It Checks:**
- **IP Addresses**
 - **Source IP** (where the packet is coming from)
 - **Destination IP** (where the packet is going)
- **Protocol Type**
 - Whether it's **TCP, UDP, or ICMP** (types of communication methods used in networking).
- **Port Numbers**
 - Checks the **source and destination ports** to allow or block certain services (e.g., **Port 80 for websites, Port 25 for emails**).

- **TCP Flags**
- Looks at control signals like **SYN, ACK, FIN, RST** (used to start, maintain, or end a connection).
- **ICMP Message Type**
- Used for network communication (like **ping** to check if a server is reachable).

1. Set a Security Policy

- Decide **what is allowed** and **what should be blocked** based on your network needs.

2. Define Allowed Traffic

- Create rules that allow only **necessary** connections based on:
 - **IP Address** (Who is sending/receiving?)
 - **Protocol** (Is it web traffic, email, or something else?)
 - **Port Numbers** (Which service is being used?)

3. Convert Rules to Firewall Syntax

Every firewall brand has its own **rule format**. You need to **rewrite** your security rules in a way your firewall understands.

4. Follow the "Least Privilege" Rule

Block everything by default and **only allow what is needed** for security.

- Example:
 - BLOCK ALL traffic first
 - Allow only trusted IPs and services (like web browsing or email).

5. Remove Unnecessary Services

- If you **don't need a service**, turn it off to **reduce attack risks**.
- Example: If your network **doesn't need FTP**, **block port 21**

- IP spoofing is when hackers use a fake IP address to trick the system. To stop this, routers should block suspicious IPs.
- Tiny fragment attacks break data into small pieces to hide harmful information. The solution is to either block tiny packets or put them back together before checking.
- Too many rules can slow down the firewall. To keep it fast, check the most common rules first. But security is always more important than speed.

Servers use ports **below 1024** (fixed numbers).

Clients use ports **1024 to 16383** (temporary numbers).

- **Common server ports:**

FTP: 20, 21

Telnet: 23

Email (SMTP): 25

Web (HTTP): 80

- Clients need free ports to connect. Stateless firewalls allow all traffic, which is risky.
- Stateful firewalls are safer because they only allow replies to requests.

Question?